Paper 294, Session 3: Public policy and institutional support for cooperatives

Institutional profiles under a common prudential framework

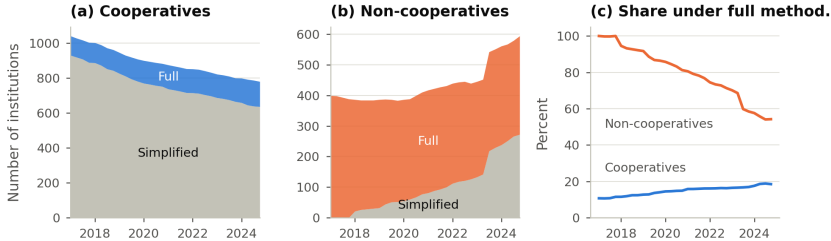
Evidence from Brazilian credit cooperatives and banks

Arthur Gomes Nery, Thiago de Oliveira Victorino and Rodrigo Lima Rangel

Organization of Brazilian Cooperatives (OCB System)

ICA CCR Research Conference, Valencia, 4 November 2026

One rulebook for two forms



The threshold

Five segments. S1 to S4 apply the full Basel methodology; small institutions may opt for a simplified method (S5). Banks are excluded from S5 at any size.

The same rules

A cooperative above the threshold, or declining the simplified option, computes capital, risk weights and provisions as a commercial bank does.

One rule apart

Cooperative acts, the operations with members, are untaxed; banks pay income tax and the social contribution. Profitability is measured before and after tax.

Data and design

Panel

IF.data, the Central Bank of Brazil's public repository. Every cooperative and bank under the full methodology, 2017Q1 to 2024Q4: 130 singular credit cooperatives, 144 banks, 7,234 institution-quarters. Central cooperatives excluded.

Inference

Standard errors clustered by institution; every coefficient confirmed by a wild cluster bootstrap. Conditional medians and quantile regressions beside the means. Cooperative status is time-invariant, so the estimates are conditional correlations.

Four specifications
(1) raw; (2) log assets and quarter effects; (3) coarsened exact matching on size; (4) common support

A ladder of comparison groups, on the regulator's categories

All non-cooperatives (483)

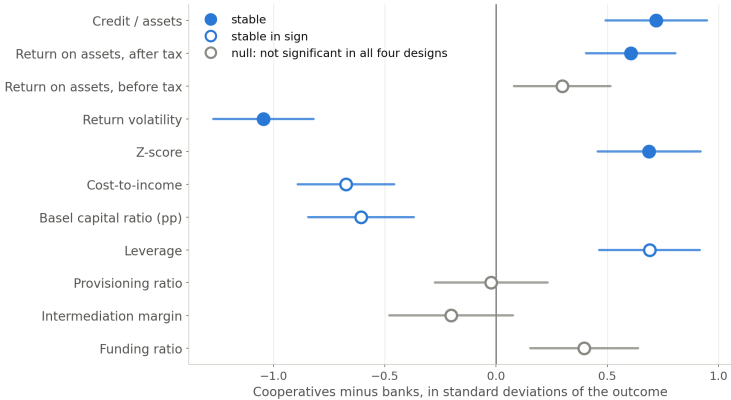
Banks (144), the primary group

Banks, matched within macro-region

Commercial banks (105)

Deposit-funded banks (94)

The profile in one picture



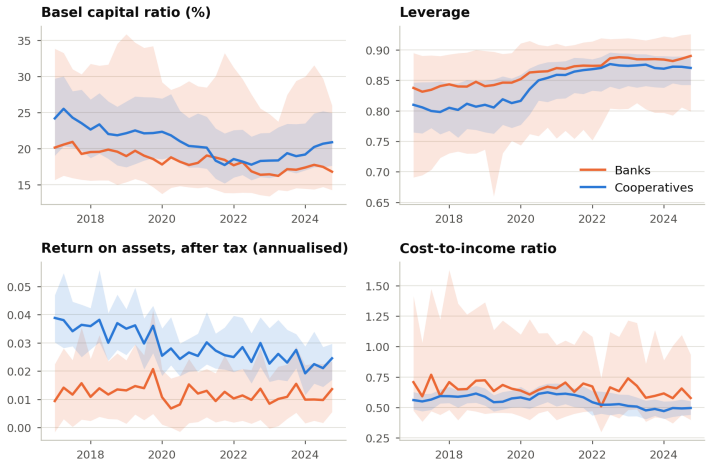
Reading

Cooperatives minus banks of comparable size, in standard deviations of each outcome. Log assets and quarter effects; 95 percent intervals.

Rule

Null unless significant with one sign in all four specifications; stable if it also survives every winsorisation rule and holds across the middle of the distribution.

Over the whole window, cooperatives are the tighter group



0.31

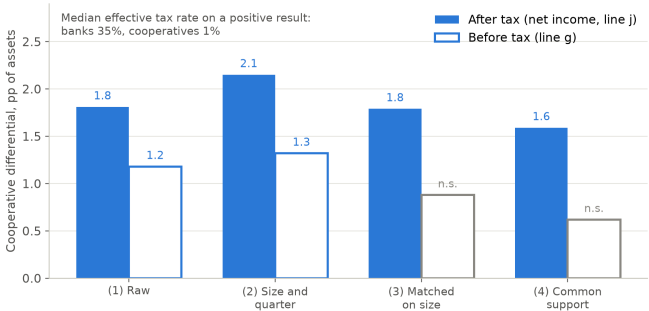
residual s.d. of the capital ratio, cooperatives over banks, at equal size

0.10

same ratio for cost-to-income

Lines are group medians, bands the interquartile range. The median cooperative is better capitalised than the median bank for most of the window. Dispersion ratios from tables/t8.

Profitability: the operating return and the tax wedge



After tax: stable

About two percentage points of assets more than comparable banks, in every specification: the return a cooperative retains as capital.

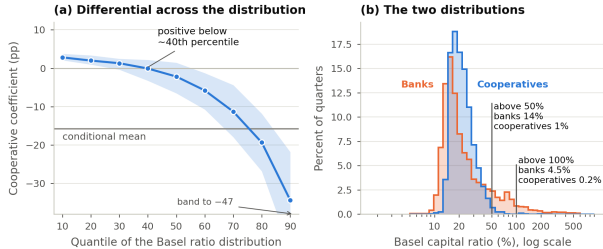
Before tax: null at the mean

Positive at the median (1.4 pp, interval excludes zero; tables/t7). A high-earning bank tail pulls the mean towards zero.

The wedge

The pre-tax gap under controls is three fifths of the after-tax gap, so the tax exemption accounts for about two fifths.

Capital: a modest gap in level, a large gap in dispersion



−15.8 pp

size-controlled mean gap; conditional median −2.2 pp

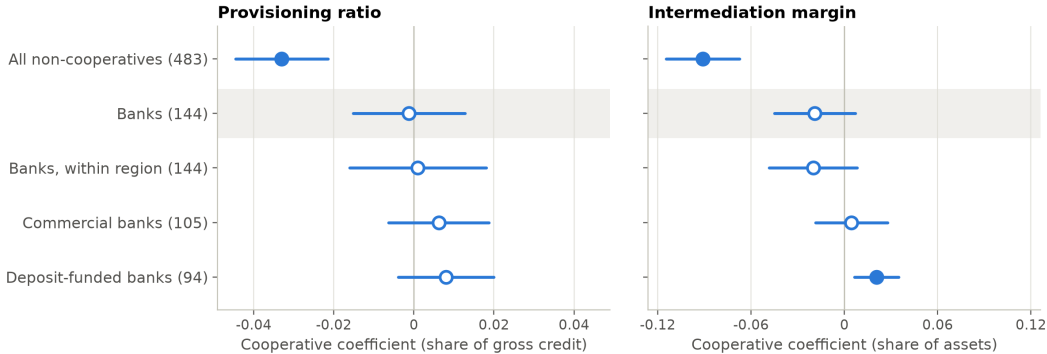
14% vs 1%

bank and cooperative quarters above a Basel ratio of 50 percent

40th

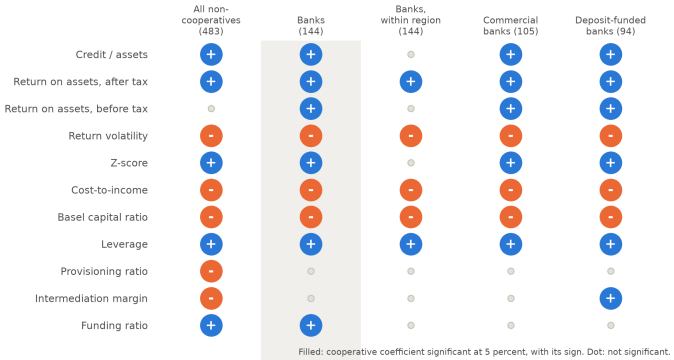
percentile where the coefficient crosses zero: positive below, negative above

Two often-cited differences vanish against banks



Against all non-cooperatives, cooperatives appear to provision less and earn thinner margins. Against banks both coefficients sit at zero. The broad group holds 158 institutions with no credit and no funds raised, and non-banking lenders whose provisioning ratios are nearly three times those of banks and cooperatives.

What survives against which comparison group



Holds on every rung

After-tax return, cost-to-income, return volatility, capital and leverage.

Holds nationally, open within region

Credit intensity: nearly a fifth of assets more than banks of the same size; zero within macro-region (effective control sample of about eighteen banks).

Only against the broad group

Lower provisioning and thinner margins.

What it means

For proportionality

A benchmark calibrated on the average institution describes the lending cooperative poorly, since cooperatives do not inhabit the upper region of that distribution. Two margins beyond size: integrated networks and their backstops, and the calibration itself. Lighter requirements do not follow automatically.

For the tax debate

The exemption of cooperative acts appears in the retained return, the only capital a cooperative can build, and accounts for about two fifths of the after-tax differential. At the banks' effective rate, cooperative capital would grow about three tenths more slowly.

For the movement

Cost at the median and return stability hold against every definition of a bank. Profitability holds for the retained return and, before tax, at the median. More conservative provisioning does not hold. The territorial claim remains open.

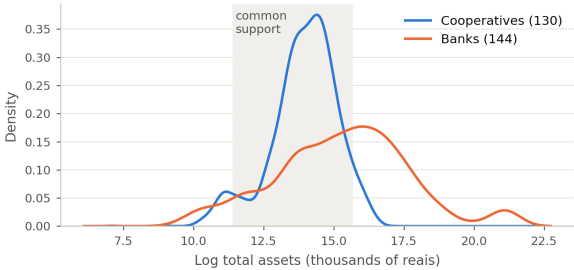
Three analyst choices decided whether headline results existed: the comparison group, reading cumulative income fields as quarterly flows, and the matched estimand. Code and pipeline are public.

Supplementary

Methods and details

For questions

Supplementary: who is in the sample



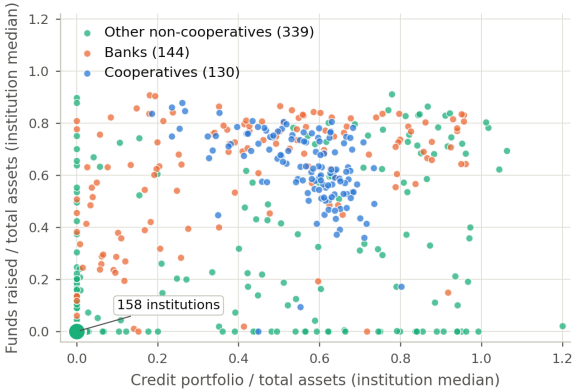
Common support

Ninety percent of cooperative observations fall inside the common support in log assets, against 48 percent of bank observations. Matching and trimming bring the standardised difference from −0.72 to about zero.

Classification

On the Central Bank's consolidation type: b3S singular cooperative, b3C central (excluded). Each institution takes its modal type, so status is time-invariant; classification by name would place the sector's own banks on the cooperative side.

Supplementary: who is in the broad comparison group



339 institutions outside the bank rung

Of these, 158 have a median of zero credit and zero funds raised, which makes the credit and funding differentials mechanical; the provisioning gap comes from non-banking lenders.

No rung defined on lending

Selecting peers on an outcome would build the credit result in. The broad group is reported because its divergence from banks is itself a result.

Supplementary: two measurement corrections

Income fields accumulate within the semester

Brazilian institutions close balance sheets in June and December, and IF.data income fields reset at each date. Flows are de-cumulated and annualised, which scales the return coefficient by about 2.7 relative to reading the cumulative field as a quarterly rate; the de-cumulated medians match the Central Bank's published figures.

The intermediation result is net of provisions

The expense subtotal contains the loan-loss provision line, so the reported result partly reproduces the provisioning outcome under another name. The margin is computed gross of provisions, and the cost-to-income denominator likewise.

Provisioning is an accounting variable

Stock of provisions under Resolution 2.682/1999 over gross credit; minimum provisions follow the AA to H rating grid, independent of ownership. A weak proxy for realised delinquency.

Supplementary: the cooperative differential against banks

Outcome	(1) Raw	(2) Controls	(3) CEM	(4) Trim	Median (2)	ρ	Tier
Basel capital ratio (pp)	-8.19***	-15.77***	-17.50***	-14.21***	-2.18	1.73	stable in sign
Leverage	0.0561**	0.1220***	0.1046***	0.0813***	0.0283	1.45	stable in sign
Return on assets, after tax	0.0181***	0.0215***	0.0179***	0.0159***	0.0171	0.88	stable
Return on assets, before tax	0.0118***	0.0132**	0.0088	0.0062	0.0144		null
Cost-to-income ratio	-0.6025***	-0.7923***	-0.8385***	-0.7522***	-0.1796	1.25	stable in sign
Credit portfolio / assets	0.1654***	0.1881***	0.1701***	0.1780***	0.2448	1.08	stable
Provisioning ratio	0.0020	-0.0012	0.0010	0.0009	0.0236		null
Intermediation margin	-0.0003	-0.0190	-0.0107	-0.0171	0.0203		null
Funding ratio	0.0461	0.0887**	0.0536	0.0330	-0.0247		null
Return on assets volatility	-0.0110***	-0.0189***	-0.0175***	-0.0170***	-0.0133	1.55	stable
Z-score	4.20***	6.32***	6.14***	6.20***	7.58	1.48	stable

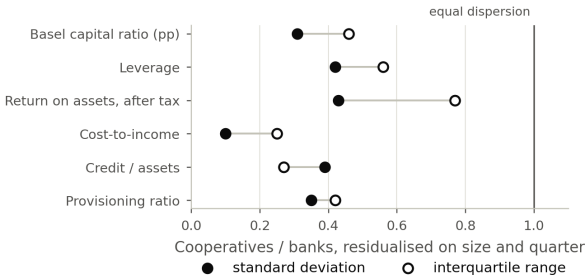
Standard errors clustered by institution, HC3 for the last two rows; * p<0.05, ** p<0.01, *** p<0.001, confirmed by a wild cluster bootstrap. Median (2): conditional median of column (2). rho = |b_trim|/|b_raw|. Tier: see Section 3.4.

Supplementary: quantile regression coefficients

Outcome	q_{10}	q_{25}	q_{50}	q_{75}	q_{90}	Mean
Basel ratio (pp)	2.81	1.61	-2.18	-14.23	-34.37	-15.77
	[2.06, 3.75]	[0.19, 2.76]	[-6.52, 1.43]	[-22.45, -8.20]	[-47.28, -21.82]	[-21.93, -9.62]
Leverage	0.294	0.117	0.028	-0.009	-0.023	0.122
	[0.194, 0.370]	[0.069, 0.174]	[0.003, 0.062]	[-0.021, 0.008]	[-0.037, -0.007]	[0.082, 0.162]
Return on assets, after tax	0.044	0.026	0.017	0.010	-0.006	0.021
	[0.032, 0.062]	[0.020, 0.030]	[0.014, 0.021]	[0.005, 0.015]	[-0.017, 0.004]	[0.014, 0.029]
Return on assets, before tax	0.052	0.028	0.014	-0.003	-0.029	0.013
	[0.038, 0.066]	[0.023, 0.035]	[0.009, 0.019]	[-0.011, 0.005]	[-0.046, -0.016]	[0.004, 0.023]
Cost-to-income	0.110	0.006	-0.180	-0.647	-1.756	-0.792
	[0.009, 0.151]	[-0.051, 0.067]	[-0.277, -0.107]	[-0.870, -0.438]	[-2.822, -1.158]	[-1.047, -0.538]
Credit / assets	0.383	0.424	0.245	-0.019	-0.170	0.188
	[0.343, 0.428]	[0.370, 0.466]	[0.146, 0.363]	[-0.085, 0.069]	[-0.219, -0.111]	[0.129, 0.248]
Provisioning ratio	0.030	0.030	0.024	0.001	-0.054	-0.001
	[0.026, 0.033]	[0.026, 0.034]	[0.018, 0.029]	[-0.016, 0.015]	[-0.093, -0.017]	[-0.015, 0.013]

Quantile regressions on the cooperative dummy, log assets and quarter effects; the last column is OLS. Brackets: 95 percent institution-bootstrap intervals (199 draws) and the clustered interval for the mean.

Supplementary: dispersion at equal size



Residuals from log assets and quarter effects

Cooperative over bank residual s.d.: capital 0.31, leverage 0.42, after-tax return 0.43, cost-to-income 0.10, credit 0.39, provisioning 0.35 (tables/t8); generalised variance ratio over seven outcomes 0.08.

Reading

Cooperatives are the more homogeneous group on every outcome. Endowment capital predicts a narrow band: no reason to hold capital far above what lending requires, and no outside investor to supply it.

Supplementary: the bank upper tail sits among banks reporting individually

Group	Institutions	Quarters	Mean	Median	p_{90}	Share $>$ 50%
Banks, conglomerate consolidations	87	2,604	22.7	16.8	39.2	6.4
Banks, individual	57	1,580	44.5	24.5	109.1	26.4
Cooperatives (all individual)	130	3,050	22.8	20.5	32.9	1.0

Basel ratio, winsorised analysis sample (tables/t11, panel A). Against individually reporting banks the controlled capital gap is −21.4 pp; against prudential conglomerates, −5.4 pp. Against deposit-funded banks the share of bank-quarters above 50 percent falls to 7.5 percent and the mean gap roughly halves while remaining significant.

Supplementary: heterogeneity by cooperative system

System	Inst.	Treated	Basel (pp)	Leverage	ROA	Provisioning
Sicredi	106	106	-17.35***	0.1341***	0.0219***	-0.0008
Sicoob	17	17	-10.98*	0.0847*	0.0164*	-0.0007
Independent	6	5	6.90	-0.0313	0.0287*	-0.0165
Cresol	1	–	fewer than five institutions, not estimated			

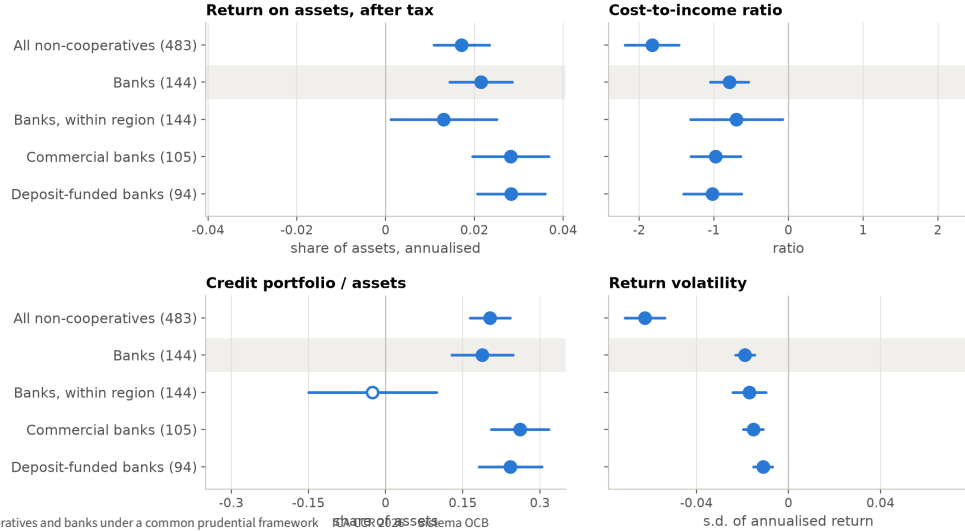
Controlled specification against the bank group. Stars from the restricted wild cluster bootstrap: * p<0.05, ** p<0.01, *** p<0.001. Treated: cooperative clusters with a usable observation; with five, the smallest attainable bootstrap p-value is about 0.03. Sicoob's capital coefficient has a bootstrap p-value of 0.019. The independent return-on-assets coefficient has an asymptotic p-value below 0.001 and a bootstrap p-value of 0.033.

Supplementary: before and after the 2022 capital rules

Outcome	Full sample	2017Q1–2021Q4	2022Q1–2024Q4
Basel capital ratio (pp)	-15.77***	-16.90***	-13.96***
Leverage	0.1220***	0.1337***	0.1033***
Return on assets, after tax	0.0215***	0.0242***	0.0173***
Return on assets, before tax	0.0132**	0.0145*	0.0111*
Cost-to-income ratio	-0.7923***	-0.8533***	-0.6927***
Credit portfolio / assets	0.1881***	0.2046***	0.1629***
Provisioning ratio	-0.0012	-0.0082	0.0104
Intermediation margin	-0.0190	-0.0251	-0.0093
Funding ratio	0.0887**	0.1128***	0.0508
Observations	7,234	4,436	2,798
Cooperatives	130	112	120
Banks	144	141	134

Controlled specification, cooperatives against banks, on the full sample and on each subperiod; the boundary is 3 January 2022, when Resolutions CMN 4.955 and 4.958 took effect. Standard errors clustered by institution. Bootstrap p-values (1,999 replications) agree with the clustered stars in every cell.

Supplementary: the ladder, outcome by outcome



Supplementary: credit intensity within and across regional markets

Nationally

Matched on size alone, cooperatives allocate nearly a fifth of assets more to credit than comparable banks, significant at one percent in every specification; at the conditional median, about a quarter.

Within macro-region

Matched also on region, the differential is small and indistinguishable from zero under any coarsening scheme. Cooperatives concentrate in the South, banks in the Southeast; the effective control sample is about eighteen banks.

Which estimate is the parameter

If cooperatives concentrate where they do for reasons internal to member-based finance, region is a mediator and the national estimate is the parameter. If regional market structure is unrelated to form, region is a confounder and credit intensity is a null. The assumption is untestable here, so both are reported.

Next step

A market measure finer than macro-region, and IF.data's credit-modality modules, which carry rural credit.

Supplementary: selection into the full methodology

Who adopts

The cooperatives observed are the largest in the country, about a tenth of all cooperatives in the panel. Fifty-one are observed under the simplified methodology before adopting the full one.

What it means

The estimates describe the adopters. The wider cooperative population would, if anything, show larger credit and profitability differentials and a smaller efficiency one.

Adopters against never-adopters, pre-adoption quarters

No difference in leverage; adopters were less credit-intensive (-0.068, $p = 0.002$), marginally less profitable and marginally leaner.

No regime effect

A within-institution design around adoption fails to identify one: adoption follows growth, and pre-adoption trends run through the adoption date without a break.

Supplementary: literature and expectations

Expected from theory

Less regulatory capital in a narrower band (endowment capital, Fonteyne 2007; property-rights problems, Cook 1995); a higher share of assets in credit; lower return volatility (Hesse and Čihák 2007). No sign committed on profitability and efficiency; no difference expected in provisioning or margin, which follow a regulatory grid.

Against the cross-country evidence

Agreement on lower capitalisation at comparable size and lower return volatility. Disagreement on profitability: in this population the typical cooperative earns more before tax at the median, retains more after tax, and is less volatile on both measures. Part is the tax exemption; part may reflect a population of the largest cooperatives.

Brazilian strand

Bressan et al. 2011, Carvalho et al. 2015, Silva et al. 2023 (PEARLS on S4 credit unions, who proposed the comparison with banks of the same segment).

Supplementary: limitations

Identification

Cooperative status is time-invariant, so the estimates are conditional correlations, not causal effects of form.

Population

The full-methodology restriction selects the largest cooperatives; results generalise to that subpopulation.

Measures

Provisioning is an accounting variable. Stability outcomes are institution-level statistics on a subsample with at least eight quarters.

Networks

Most systems are too thin for separate inference. The backstop hypothesis needs consolidated network data that IF.data does not carry; centrals held about six percent of the singulars' regulatory capital at end-2024, so consolidation would move the differentials by far less than their size.

Markets

The mediator question needs a market measure finer than macro-region.